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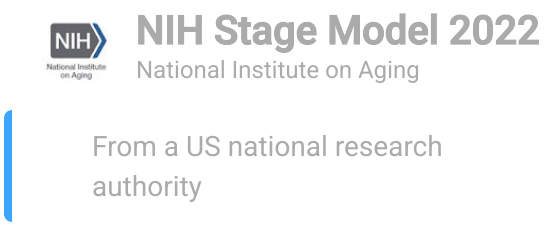
NIH Stage Model for Behavioral Intervention Development

The Stage Model is a model of behavioral intervention development composed of six stages: basic science (Stage 0), intervention generation, refinement, modification, and adaptation and pilot testing (Stage I); traditional efficacy testing (Stage II); efficacy testing with real-world providers (Stage III); effectiveness research (Stage IV) and; dissemination and implementation research (Stage V). Examination of mechanisms of behavior change is encouraged in every stage of intervention development. Consideration of the intervention's ease of implementation is encouraged as early as possible in the intervention development process. The ultimate goal is to produce highly potent and maximally implementable behavioral interventions that improve health and well-being.

Learn about the NIH Stage Model through this training video:



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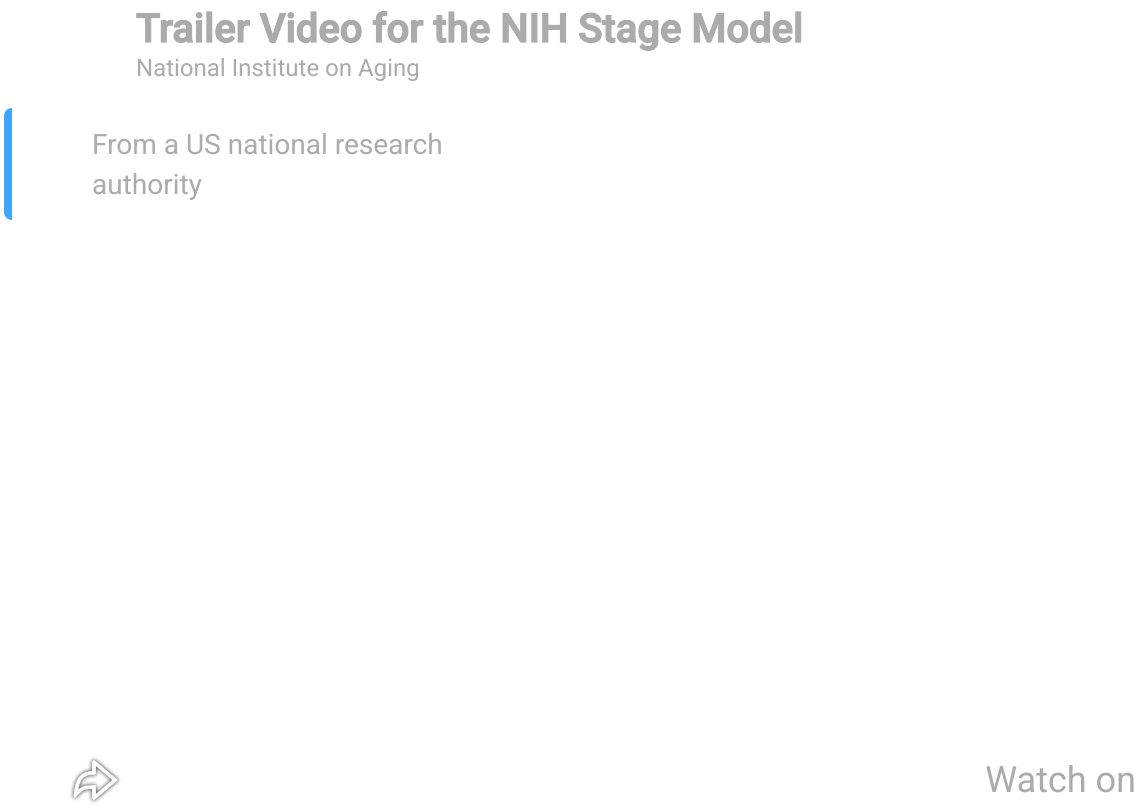
The shorter version of the training video below gives a brief overview of the NIH Stage Model:



Learn about intervention fidelity within the NIH Stage Model through this training video:

NIH Stage Model: Intervention Fidelity

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NIH Stage Model Overview

Various conceptualizations of research on intervention development share the notion of phases or stages of intervention development, and most stress the importance of translational research. The models generally agree that efficacy and effectiveness research vary along a continuum, from maximizing internal validity to maximizing generalizability. Models differ in what stages they include and in the way they number and name the stages. Models also differ in terms of the relevance, importance, and role of theory and basic research in intervention development; and in terms of the point at which they emphasize a focus on implementation.

The NIH Stage Model for Behavioral Intervention Development offers guidance on best practices for generating, testing, and implementing effective behavioral interventions that can be delivered in real-world settings. Because behavioral interventions frequently do not move beyond efficacy to effectiveness or implementation, several stages of intervention development are identified, defined, and clarified in this model to address this issue. For example, early-stage intervention development, refinement, and adaptation is broadened to include intervention modification to promote ease of implementation, and, where needed, the development of training materials (Stage I). Furthermore, two stages for different types of real-world testing are distinguished: One a hybrid efficacy-effectiveness stage to demonstrate that it is possible to administer an intervention correctly in the real-world (Stage III); another to conduct true effectiveness testing (Stage IV). Finally, the model is intended to emphasize both the scientific and practical value of determining the mechanism of action of interventions; 1) to help to create a cumulative, progressive field and; 2) to help identify principles of behavior change that can be imparted to those who are delivering interventions. This emphasis on mechanisms facilitates the ability to operationalize personalized

interventions, tailored for different characteristics of individuals, couples, families, for broad range of behaviors and across settings.

The NIH Stage Model is an iterative, recursive, multidirectional model of behavioral intervention development. This model asserts that intervention development is not complete until an intervention reaches its maximum level of potency *and* is implementable with a maximum number of individuals in the population for which it was developed. In this model basic researchers, intervention developers, and community-oriented intervention and implementation researchers all have a significant role to play in every stage of developing potent and implementable interventions.

Stages of Intervention Development

- **Stage 0** involves basic science that occurs *prior to* intervention development, but is relevant (ultimately translatable) to intervention development. Another type of basic science research—research on mechanisms of change—is an integral part of all other stages of intervention development. Such research involves asking basic science questions about behavior change *within the context of* intervention development studies. Questions of mechanisms of behavior change are relevant to every Stage of behavioral intervention development.
- **Stage I** encompasses all activities related to the creation and preliminary testing of a new behavioral intervention. Stage I can include the generation of new behavioral interventions as well as the modification, adaptation, or refinement of existing interventions (Stage IA), and it culminates in feasibility and pilot testing (Stage 1B). Stage I can also include modification of an intervention for the purpose of making it more easily implementable. In addition, also in the service of increasing implementability, Stage I can include, as appropriate and necessary, the development of training materials. Finally, note that one can conduct Stage I studies in research settings, with research providers and research subjects; or in community settings with community providers or caregivers.
- **Stage II** (Pure “Efficacy”) research consists of experimental testing of promising behavioral interventions *in research settings, with research-based providers.*
- **Stage III** (Real World “Efficacy”) research consists of experimental testing of promising behavioral interventions *in community settings, with community-based providers or caregivers*, while maintaining a high level of control necessary to establish internal validity. Some refer to this as a hybrid (efficacy-effectiveness) stage.
- **Stage IV** (“Effectiveness”) research examines empirically supported behavioral interventions in community settings, with community-based providers or caregivers, while maximizing external validity.
- **Stage V** (“Implementation and Dissemination”) research examines strategies of implementation and adoption of empirically supported interventions in community settings.

A Few Clarifications and Some Examples

- This Stage Model is *not* prescriptive. Rather, it creates a common language for writing, discussing and reviewing behavioral intervention development research (e.g., grant applications, journal articles, etc.), with the goal of ultimately leading to a more coherent, efficient, and progressive science.
- As a non-prescriptive model, it does not require that research be done in a pre-specified order: What is required is that investigators adequately justify the logic of their proposed sequence. For example, depending upon the characteristics of the intervention, Stage I may lead to Stage II or may lead directly to Stage III (e.g., if the Stage I work was done in a community setting, with community providers). It is equally plausible that Stage II, III, or IV will lead to further modification of the intervention in Stage I.
- Some sequences, however, are more easily justifiable than others. For example, a completely computerized intervention might progress quickly from Stage I to Stage III or even Stage IV effectiveness testing because of a high degree of certainty that it can be easily delivered with fidelity in the real world. On the other hand, it is not always reasonable to expect that a positive Stage II efficacy study will automatically or even usually lead to a

positive effectiveness study. Even after a successful Stage II efficacy study is conducted on a non-computerized intervention that requires in-person delivery, the intervention may require additional Stage I modification of the intervention and/or the training materials for the intervention before proceeding to real-world Stage III efficacy testing. Depending upon the results of the Stage III study, progression to Stage IV effectiveness testing may be warranted. Another possibility is additional Stage I work.

- Basic science questions and basic science research paradigms are integral to the entire behavioral intervention development process.
- Basic science questions and paradigms are *not limited* to research that occurs prior to intervention development. For example, research on mechanism of action (i.e., asking *how and why* a behavioral intervention works) is equivalent to asking basic science questions and using basic science paradigms within the context of applied behavioral intervention studies.
- Understanding mechanism of action and the principles behind an intervention may involve basic science expertise, but can have pragmatic effects, such as helping to: (a) Boost the effects of interventions; (b) pare down interventions to what's essential, which can make the intervention more implementable and cut cost; and (c) simplify interventions for easier transportability.
- Behavioral intervention development is incomplete until the efficacious intervention is also implementable, which in most cases means that the intervention package includes materials describing how to ensure that community providers or caregivers administer the intervention with *fidelity*. Therefore, methods for enhancing and maintaining the fidelity of intervention delivery (e.g., training and supervision materials) are an essential part of any provider-delivered behavioral intervention, and should be developed in Stage I, even after an intervention has proven efficacious in Stage II.

NIH Stage Model Resources

- The Research Centers Collaborative Network of the National Institute on Aging convened a workshop on Achieving and Sustaining Behavior Change to Benefit Older Adults in which Dr. Lisa Onken provided an in-depth overview of the NIH Stage Model. View a recording of the presentation.
- Onken, L., Carroll, K., Shoham, V., Cuthbert, B., & Riddle, M. (2014). Reenvisioning clinical science: Unifying the discipline to improve the public health. *Clinical Psychological Science*, *2*, 22–34.
- Onken, L. (2022). Implementation Science at the National Institute on Aging: The Principles of It, *Public Policy & Aging Report*, *32*(1), 39-41.

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